Overview

The goal of this task is to design an **Inertial Measurement Unit (IMU) + Magnetometer breakout board** for use in the CanSat. In this hypothetical scenario, CanSat is designing an in-flight experiment to evaluate the performance of a deployable boom. To do so, we need to accurately estimate the orientation at the tip of the deployable boom.

The breakout board will be mounted at the tip of the boom and will provide acceleration, gyroscope, and magnetic-field measurements to the main flight computer. These measurements will be used with our sensor-fusion algorithm to estimate the orientation of the boom in real time. The board should be compact and easily mounted using M2 screws.

The IMU must support a minimum **output data rate (ODR) of 400 Hz**, while the magnetometer must support a minimum **ODR of 100 Hz**. These rates are intended to provide sufficient measurement frequency for the sensor-fusion algorithm.

To accommodate the mechanical team, the PCB footprint should be **as small as reasonably practical** without significantly compromising sensor performance, mechanical integrity, or manufacturability. The board must use **2–4 M2 mechanical mounting holes** for attachment to the boom.

The sensors will communicate with the main microcontroller (MCU), so the communication interface should use a **reasonable number of MCU GPIOs** while providing sufficient bandwidth for the required sensor data.

The board will be mounted in the nosecone above a **J-class rocket motor** and must withstand peak accelerations of approximately **20 g**. It will be attached to the end of a deployable boom, remaining close to the main PCB during most of the flight and extending up to **20 cm away** during descent.

The design should consider the electrical, mechanical, and environmental requirements necessary to obtain reliable sensor measurements throughout the flight. The specific **IMU, magnetometer, board dimensions, and connector** are left to the designer.

Design Requirements

1. The board must be powered from the main PCB's regulated **3.3 V supply**.
2. The IMU must support a minimum **ODR of 400 Hz**, and the magnetometer must support a minimum **ODR of 100 Hz**.
3. The board must use **2–4 M2 mounting holes** for mechanical attachment.
4. The board should be as **compact as reasonably practical** while maintaining sensor performance and manufacturability.
5. The connector and communication interface are left to the designer but must support reliable communication over a distance of **up to 20 cm**.

6. The selected components and PCB must withstand **peak accelerations of approximately 20 g** during launch.
7. The communication interface should use a **reasonable number of MCU GPIOs** while providing sufficient bandwidth for the required sensor data.

Submission and Grading

The design will be evaluated based on:

● **Electrical design:** Component selection, power distribution, and communication interface.
● **PCB layout:** Component placement, routing, and overall layout quality. ● **Mechanical design:** Board size, mounting hole placement, and mechanical suitability. ● **Requirements compliance:** Fulfillment of the specified design requirements. ● **Design justification:** Rationale for major component and design decisions. ● **Documentation:** Completeness and quality of the accompanying design notes.

Recap

The goal of this task is to design a **compact IMU + Magnetometer breakout board** that can reliably measure the orientation of a deployable boom during flight. The final product should serve as an excellent first draft

The design should:

● Meet the specified **electrical, mechanical, and performance requirements**. ● Support the required **IMU and magnetometer measurement rates**.
● Withstand approximately **20 g of launch acceleration**.
● Communicate reliably with the main PCB over a distance of up to **20 cm**. ● Use a compact, practical PCB layout with appropriate **M2 mounting holes**. ● Demonstrate clear engineering reasoning through the accompanying **design notes**.

There is no single correct solution. The specific components, communication interface, board dimensions, and implementation details are left to you. The goal is to demonstrate your ability to **research, make engineering tradeoffs, and independently develop a practical PCB design**.

The final design should be a **strong first draft of a production-ready board**, with a schematic and PCB layout complete enough that, after appropriate review and validation, the design could be sent to a PCB manufacturer for assembly and fabrication.

Overall Notes and Guide

The most important part of this task is not simply producing a PCB that passes KiCad's design checks. You should be able to explain **why you made the major design decisions you made** and demonstrate that the final design satisfies the requirements in Section 2.

Component and Integrated Circuit Selection

For the **IMU and magnetometer**, consider factors such as:

- Required **ODR and measurement bandwidth**
- Measurement range, resolution, and noise
- Accuracy and other relevant performance specifications
- Supply voltage and current consumption
- Communication interfaces and required MCU GPIOs
- Package size, footprint, and availability
- Operating temperature range
- Mechanical and environmental specifications
- Manufacturer-recommended PCB layout
- Software, driver, and library support
- Cost and long-term availability

For **connectors**, consider the required mating connector, pitch, number of positions, current rating, mechanical strength, package size, and availability. Do not assume that a particular connector family is required; alternatives from different manufacturers may provide equivalent functionality. For example, JST-SH, JST-PH, and compatible connectors from other manufacturers may have different trade-offs in size, cost, availability, and mechanical robustness.

For **resistors and capacitors**, select appropriate values, tolerances, voltage ratings, dielectric types, and package sizes. Consider whether standard package sizes such as 0402, 0603, or 0805 are appropriate for the project's assembly capabilities and whether the required values are readily available.

When comparing components, do not rely solely on specifications shown on distributor websites. **Read the manufacturer's datasheet and relevant application notes.** Important information such as ODR limitations, operating conditions, required external circuitry, startup behaviour, pin configuration, and PCB layout requirements may only be documented by the manufacturer.

For each major IC, you should understand:

- What power supplies it requires
- What external components it requires
- How it communicates with the main MCU

- Which pins need to be configured or connected
- What PCB layout considerations the manufacturer recommends
- Any limitations that could affect the overall system design

Reference designs and evaluation-board schematics can also be useful when determining how a component should be integrated.

Consider the **system as a whole** when making component choices. For example, choosing SPI instead of I²C affects the number of MCU GPIOs, achievable data rate, bus topology, wiring, and potentially the required connector. Similarly, selecting a smaller connector or passive package may reduce PCB area but make assembly more difficult.

There is no single correct component choice for this task. If multiple components satisfy the requirements, consider the trade-offs and document why you selected your final parts.

Consider at least **three candidate IMUs and three candidate magnetometers from different manufacturers** (e.g., Texas Instruments, STMicroelectronics, and Analog Devices *other companies are fine too). Compare them based on the key design factors relevant to your application, such as performance, power consumption, interface, package, cost, availability, and software support.

Also consider alternatives for the required **connectors and passive components**, including package size, specifications, cost, and availability.

Document your comparisons in tables in your Google Doc, then select and justify the components that best meet your system requirements.

Before finalizing a component, verify that it is **actively manufactured and available through the JLCPCB Part Library** and that its cost and availability are practical for the project's manufacturing schedule.

Schematic Design

The schematic defines **how the components on the board are electrically connected**. Before beginning PCB layout, the schematic should provide a complete and unambiguous representation of the intended circuit. A well-designed schematic should allow another engineer to understand the circuit without needing to refer to the PCB layout.

Begin by identifying the major functional blocks of your design, such as the **microcontroller, sensors, power supply, communication interfaces, and connectors**. Build these blocks individually, then connect them together and add the supporting circuitry required by each component.

When designing the schematic, pay particular attention to:

- Power connections, voltage requirements, and power domains
- Decoupling and bulk capacitors
- Communication interfaces and bus configuration
- Pull-up or pull-down resistors where required
- Address, mode, and configuration pins
- Interrupt, reset, enable, and status signals
- Connector pin assignments
- Protection circuitry where appropriate
- Unused pins and their required treatment

The manufacturer's **datasheet and recommended application circuit** should be your primary reference when determining how an IC should be connected. Check the manufacturer's documentation for required external components, pin configurations, operating conditions, startup requirements, and special considerations.

Keep the schematic **organized and readable**. Group related circuitry into functional blocks, use clear and consistent net names, and maintain a logical signal flow. Use hierarchical or global labels where appropriate rather than drawing unnecessarily long connections across the schematic.

As you build the schematic, consider how each design decision affects the rest of the system. For example, selecting I²C versus SPI affects the required MCU pins, pull-up resistors, achievable data rate, bus topology, and potentially the connector pinout. Similarly, the selected power architecture determines which voltage rails are available to the other components.

Before moving on to PCB layout, perform a complete schematic review. Verify that:

- Every component has been selected appropriately.
- Every IC pin has been intentionally connected or appropriately left unused. ● All power connections and voltage levels are correct.
- Required decoupling and supporting components are present.
- Communication interfaces are correctly connected and configured.
- Connector pin assignments are correct.
- Address, configuration, interrupt, reset, and enable pins are handled correctly. ● Component values and footprints are specified.
- Electrical Rules Check (ERC) has been performed and reviewed.
- Any intentional ERC warnings or exceptions are understood and documented.

Before beginning PCB layout, review your schematic against the **manufacturer's datasheets, application notes, and the system requirements**. Pay particular attention to power connections, IC pin assignments, communication interfaces, connector pinouts, and required supporting circuitry. It is significantly easier to identify and correct an electrical design error in the schematic than after the PCB has been routed.

PCB Layout

PCB layout is more than simply routing the connections from the schematic. A good layout considers **electrical performance, mechanical constraints, sensor performance, manufacturability, and assembly** as a single system. **Component placement should be planned before routing**, and important placement decisions should be based on the requirements and manufacturer recommendations identified during component selection.

Mechanical Layout

Begin by establishing the **board outline, mounting holes, keep-out areas, and connector locations**. Consider how the board will physically attach to the rest of the system and ensure that mounting hardware, connectors, and other interfaces remain accessible.

Keep the board reasonably compact, but do not reduce its size at the expense of **sensor performance, manufacturability, mechanical strength, or adequate routing space**. Leave sufficient clearance around mounting holes, connectors, and other mechanically constrained areas.

Component Placement

Place components according to their **electrical and functional relationships**. Components that interact closely should generally be placed near one another to minimize unnecessary trace length. For example, decoupling capacitors should be placed immediately adjacent to the power pins they support.

Consider the complete signal and power flow through the board when determining placement. Avoid placing components solely to make routing convenient if doing so creates problems for power integrity, thermal performance, sensor measurements, or mechanical integration.

Sensor Placement

Sensor placement is particularly important for boards containing an **IMU or magnetometer**. For the magnetometer in particular, keep potential magnetic sources as far away as practical and consider the current paths on the PCB, not just the physical location of components. For the IMU, consider the mechanical coupling between the sensor and PCB, as well as the orientation of the sensor relative to the intended coordinate system.

Power and Grounding

Place decoupling capacitors as close as practical to the corresponding IC power pins, using short connections with a suitable return path to ground. Follow the manufacturer's recommended layout for sensitive components and power circuitry.
Use an appropriate **ground-plane strategy** and consider the return path of every important signal. Avoid unnecessarily splitting or interrupting ground planes beneath signals when doing so would create poor return paths.

Keep noisy or high-current circuitry away from sensitive sensors and analog circuitry where practical. Consider both conducted and radiated sources of interference when placing power supplies and other potentially noisy components.

Layer Stackup

The PCB layer stackup should be chosen based on the **electrical, mechanical, and manufacturing requirements** of the design. For a small sensor board, a **4-layer PCB is generally a good starting point** because it provides dedicated ground and power planes while making signal routing significantly easier.

A typical 4-layer stackup could be:

Layer	Purpose
L1 – Top	Components and primary signal routing
L2 – Ground	Continuous ground plane
L3 – Power / Signal	Power distribution and secondary signal routing
L4 – Bottom	Secondary signal routing and components

Keeping a relatively continuous ground plane directly beneath the primary signal layer provides good signal return paths and can help reduce noise and EMI. Avoid unnecessarily splitting the ground plane, particularly underneath sensitive or high-speed signals.

When selecting the stackup, consider:

- Number of routing layers required by the circuit
- Need for continuous ground and power planes
- Signal integrity and return paths
- Board thickness and mechanical requirements
- Copper thickness
- Manufacturer capabilities and available stackups
- Cost and availability
- Required trace widths and controlled impedance, if applicable

Do not assume that every PCB manufacturer uses the same layer stackup. Check the manufacturer's available stackups and design rules before finalizing the board. The physical distance between layers affects trace impedance, capacitance, and return paths, so the stackup should be considered when establishing routing rules.
Suggested Starting Point: Unless the design has unusual requirements, **consider starting with a 4-layer stackup with a dedicated ground plane directly beneath the component/signal layer**. Evaluate whether a 2-layer board is sufficient or whether additional layers are justified by routing

density or electrical requirements.

Routing

Route signals with consideration for **signal integrity, return paths, current capacity, and manufacturability**. Keep important connections reasonably short and direct where practical.

Use appropriate trace widths, clearances, vias, and other design rules for the selected PCB manufacturer. Higher-current power connections may require wider traces or copper pours, while sensitive signals may benefit from being kept away from noisy or high-current paths.

Avoid routing unnecessarily close to sensitive sensor circuitry or creating routing configurations that could introduce unwanted electrical or magnetic interference.

Manufacturing

Design the PCB around the capabilities of the **intended manufacturer and assembly process**. Configure KiCad's design rules according to the manufacturer's published capabilities rather than relying solely on KiCad's defaults.

Verify that footprints correspond to the **actual component packages specified in the datasheets**. Pay particular attention to pad dimensions, exposed pads, pin-1 orientation, connector footprints, and mechanically constrained components.

Avoid unnecessarily small or difficult-to-manufacture features when they do not provide a meaningful benefit. When possible, favour commonly available package sizes, standard design rules, and components that are readily available through the selected manufacturer.

Final Layout Review

Before considering the PCB complete:

- Run the **PCB Design Rules Checker (DRC)** and resolve or review all violations.
- Verify critical footprints against the component datasheets.
- Verify connector footprints and pin assignments.
- Verify mounting-hole dimensions, locations, and clearances.
- Check component orientation, polarity, and pin-1 markings.
- Check power and ground connectivity.
- Verify sensor placement and orientation.
- Inspect critical traces, vias, and current paths.
- Check clearances around mechanical and electrical keep-out areas.
- Inspect the completed board in **3D**.
- Review the layout against the system requirements and constraints established earlier in the guide.

Passing DRC does not necessarily mean that the PCB is correct. KiCad can verify many geometric and electrical rules, but it cannot determine whether a sensor was placed in a poor location, a connector

is mechanically inaccessible, a footprint is incorrect, or a layout decision will negatively affect sensor performance.

Layout Review: Before finalizing the PCB, perform a deliberate review of the placement and routing rather than relying solely on DRC. Trace important signals from their source to their destination, inspect power and ground paths, and verify that the physical layout satisfies the requirements identified during the schematic and component-selection stages.